

Functional Requirements Document (FRD)

1. Functional Overview

The solution will provide a reporting layer and dashboard requirements for analyzing loan applications, decisions, turnaround time and operational bottlenecks.

2. Functional Requirements

- FR-001: Load and expose application records with a unique application identifier.
- FR-002: Support date-based filtering and monthly trend reporting.
- FR-003: Calculate application counts by decision status.
- FR-004: Calculate approval rate as approved applications divided by eligible decided applications.
- FR-005: Calculate rejection rate using the same documented eligible population.
- FR-006: Calculate average and median turnaround time where timestamps are available.
- FR-007: Analyze applications by loan type and region.
- FR-008: Analyze applications by customer segment.
- FR-009: Identify stage-level processing delays.
- FR-010: Provide KPI cards for application volume, approval rate, rejection rate and turnaround time.
- FR-011: Provide trend, comparison and distribution visuals.
- FR-012: Provide slicers for date, region, loan type and decision status.
- FR-013: Support drill-down from summary to relevant analytical dimensions.
- FR-014: Apply documented data-quality validation rules.
- FR-015: Maintain traceability to BRD requirements and UAT cases.

3. KPI Definitions

- Application Volume = count of unique in-scope applications.
- Approval Rate = $\text{approved applications} / \text{eligible decided applications} \times 100$.
- Rejection Rate = $\text{rejected applications} / \text{eligible decided applications} \times 100$.
- Pending Rate = $\text{pending applications} / \text{total in-scope applications} \times 100$.
- Turnaround Time = decision timestamp minus application submission timestamp.

4. Non-Functional Requirements

- Performance: dashboard interactions should be responsive for the agreed dataset size.
- Security: access must follow business role requirements.
- Data Quality: mandatory identifiers and timestamps must be validated.
- Usability: KPI labels and definitions must be understandable to business users.
- Auditability: calculation rules and requirement versions must be documented.

5. Acceptance Criteria Examples

- Given a selected month, the dashboard displays only applications within that month.
- Given a loan type filter, KPI cards and visuals recalculate for that loan type.

- Approval rate excludes applications that do not meet the documented decided-status population.
- Turnaround time is calculated only where required timestamps are valid.
- UAT evidence can be traced to the relevant requirement.

Portfolio Disclosure

Functional requirements are simulated/illustrative and intended to demonstrate BA documentation skills.